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The effects of direct and indirect contact on prejudice: 2007 and 2017 results among Greek Cypriots and Turkish Cypriots

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ABSTRACT
In post-conflict and segregated societies, overcoming high prejudice between groups is essential for peaceful coexistence. In this paper, we explore the roles of direct contact (face-to-face contact and cross-group friendship) and indirect contact (extended cross-group friendship) in reducing prejudice between Greek Cypriots and Turkish Cypriots living in Cyprus. Moreover, this study further explores whether the effects of direct and indirect contact vary between Turkish Cypriots and Greek Cypriots, as well as across a ten-year period. To explore these issues, we analyze data from different representative samples of Greek Cypriots and Turkish Cypriots at two different time points (years 2007 and 2017). The results suggest that all three forms of contact have significant effects on prejudice. However, they also exhibit some variations by community (but not year of data collection) in the effects of direct (but not indirect) contact on prejudice.

KEYWORDS Direct contact; indirect contact; prejudice; Greek Cypriots; Turkish Cypriots

Introduction

23 April 2003 was a historic date for Cyprus. Twenty-nine years after the events of 1974 that brought complete geographical separation between Greek Cypriots (GCs) and Turkish Cypriots (TCs), members from both communities could finally resume contact after a handful of checkpoints were opened. This partial lifting of travel restrictions offered a unique opportunity for social scientists to explore in real-life conditions the potentially beneficial effects of intergroup contact on reducing prejudice, an important element of the ongoing efforts to reach a viable and sustainable solution to the Cyprus issue.

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While the opening of checkpoints has created many opportunities for face-to-face contact in Cyprus since 2003, opportunities for such contact are limited since most Cypriots still live in fully segregated conditions (Husnu *et al.* 2018; Ioannou 2009; Psaltis and Lytras 2012). As shown in prior research, these opportunities also depend on the size of the groups in question (Blau and Schwartz 1984). Moreover, this form of contact is less effective for minorities than for majorities (Pettigrew and Tropp 2006), and it might not be pursued at all due to psychological barriers such as intergroup anxiety (Stephan and Stephan 2000; Zezelj *et al.* 2015). Consistent with the contact hypothesis by Allport (1954), most prior research has focused on the frequency or quantity of face-to-face contact (Pettigrew and Tropp 2006). While quantity measures are mute about the quality of contact, merely noting the time one meets with or talks to outgroup members, some other research has focused on more 'meaningful' and high-quality interactions across different groups, namely direct cross-group friendships (Turner *et al.* 2007; Davies *et al.* 2011).

Pettigrew (1997) described friendship as having a special importance in intergroup relations 'because it involves contact over time and across many situations, through which group members develop meaningful, close relationships under conditions that facilitate improved attitudes (e.g. cooperation, equal status; see Allport 1954)' (Davies *et al.* 2011: 333). Indeed, prior research has shown that cross-group friendship is associated with more positive intergroup attitudes (see the meta-analysis by Davies *et al.* 2011). In order to capture both frequent and high-quality contact, this study uses both measures for direct contact (i.e. quantity of face-to-face contact, and direct cross-group friendship). When direct contact is not possible, indirect contact, which does not require face-to-face interactions and eliminates any intergroup anxiety participants might face during direct contact, is found to promote better intergroup relations (Wright *et al.* 1997). One form of indirect contact, extended cross-group friendship, was found to lead to more positive intergroup attitudes and more positive intergroup behavioral intentions (see the review by Vezzali *et al.* 2014).

In this paper we intend to compare findings from two large-scale representative sample surveys of Greek Cypriots and Turkish Cypriots. The first survey was undertaken in 2007, a few years after the opening of the checkpoints and just before the establishment of several bi-communal technical committees by the two community leaders – part of the process to establish confidence-building measures 'aimed at not only improving the

everyday lives of Cypriots, but also at encouraging and facilitating greater interaction among them' (UNSG 2017). The second survey took place 10 years later, in 2017, to explore the effects of both direct and indirect contact on prejudice reduction between the two communities. Using data from both year 2007 and 2017, this study asks the following four research questions: (1) Does direct contact (quantity of face-to-face contact and direct cross-group friendships) lead to lower prejudice between Greek Cypriots and Turkish Cypriots? (2) Does indirect contact (extended cross-group friendship) lead to lower prejudice between Greek Cypriots and Turkish Cypriots? (3) Do the effects of direct and indirect contact on prejudice differ between Greek Cypriots and Turkish Cypriots? (4) Do the effects of direct and indirect contact on prejudice differ between the 2007 and 2017 survey data?

The contribution of this paper is twofold. First, it is one of the few studies ever performed in Cyprus analyzing representative samples from both communities in parallel (Yucel and Psaltis 2020; McKeown and Psaltis 2017; Psaltis *et al.* 2019). Second, it introduces a novel time perspective to contact-related research in Cyprus, by comparing 2007 data with 2017 data. To our knowledge, no other published research has made such comparisons, in any geographical context.

The Cyprus context

Cyprus was part of the British Empire during the 1950s. During that time, Greek Cypriots (who constituted 82% of the population) began to seek a union with Greece. This, however, was strongly opposed by the Turkish Cypriots (the other 18% of the population), who then started their own movement to partition Cyprus in two. This conflict led to violent intercommunal clashes, internal displacement of the population, and increased segregation. Ultimately, a coup in 1974 that aimed at unifying Cyprus with Greece prompted a military invasion by Turkey, resulting in ethnic cleansing, mass displacements, and the division of the island into two ethnically homogeneous areas, separated by a UN-patrolled cease fire line. Later, in 1983, the Turkish Cypriots in North Cyprus issued a Unilateral Declaration of Independence, establishing a breakaway state (see Papadakis 2008) recognized only by Turkey.

For almost 30 years, the occupied north was sealed off from the southern region (inhabited by the GC community), but opportunities for contact between GCs and TCs changed dramatically in 2003 when travel restrictions between the north and south were lifted. Since then, it

is estimated that about 70% of the residents from both communities has crossed at least once to the other side, although only about a third of GCs and TCs reported regular contact with members of the other community (Psaltis 2015; Yucel and Psaltis 2020).

Theoretical framework

This study uses Allport's (1954) contact hypothesis. Higher contact, under four conditions – equal status, common goals, acquaintance potential, and support by authorities or ingroup norms – is expected to lead to the reduction of existing prejudice, a trend which has been largely supported in prior research (see the meta-analysis by Pettigrew and Tropp 2006). One ramification of the original contact hypothesis is that closer forms of contact, such as cross-group friendships, are expected to be particularly effective at improving intergroup relations through mechanisms such as the reduction of prejudice. Pettigrew (1997, 1998) argued that one way to improve intergroup contact theory is for researchers to focus on these unique forms of contact. Cross-group friendship is defined as 'friendships that form between members of different groups who are in direct contact with one another' (Turner *et al.* 2007: 213). Pettigrew (1998), while he supported Allport's (1954) model of four optimal conditions of contact, added a fifth condition: intergroup contact, he argued, should also facilitate opportunities for friendships between members of different groups. Cross-group friendship is found to promote more positive intergroup attitudes (see the meta-analysis by Davies *et al.* 2011; Pettigrew 1997).

An equally important theoretical development in this field is the exploration of indirect forms of contact[1] and their potential prejudice reduction effects (Brown and Paterson 2016). This study focuses on one form of indirect contact: extended contact, specifically extended cross-group friendship. Extended cross-group friendship is defined as 'the mere knowledge that other ingroup members have cross-group friends' (Turner *et al.* 2007: 213). The extended contact hypothesis is based on

[1]Indirect contact refers to a series of experiences that include: 1) Knowledge of another ingroup member having contact (or friendship) with an outgroup member (Wright et al., 1997), 2) Observation of ingroup and outgroup member having real-life direct contact (i.e., vicarious contact) (Mazziotta et al., 2011; Ioannou et al., 2018) 3) Media-based contact (consuming TV, films, radio shows, or documentaries featuring ingroup and outgroup members interacting) (Paluck, 2007), 4) Computer-mediated intergroup contact, like having friendships on social media (Zezelj et al., 2017) or interacting through chat platforms, and 5) imagined contact, which refers to the simulation of a contact experience in one's imagination (Husnu and Crisp 2010).

the fact that merely being aware of intergroup friendships between a member of one's own group and another group can also improve intergroup attitudes (Wright *et al.* 1997). Prior research showed that extended contact is positively associated with intergroup attitudes (see the meta-analysis by Zhou *et al.* 2019).

Overall, using the contact and extended contact hypotheses as a foundation, this study argues that direct and indirect contact are both expected to reduce prejudice between Greek and Turkish Cypriots.

Intergroup contact and prejudice

This study considers two forms of direct contact (face-to-face contact and cross-group friendships) and one form of indirect contact (extended cross-group friendship). A widely cited meta-analysis of 515 prior contact studies reported that face-to-face contact did lead to prejudice reduction (Pettigrew and Tropp 2006). While most prior research focused on face-to-face contact, another form of direct contact is direct cross-group friendship. Direct cross-group friendship is considered to be one of the most important forms of prejudice-reducing contact, as it facilitates powerful affective processes like reduced intergroup anxiety, reciprocal self-disclosure, and the promotion of empathy and perspective-taking (Pettigrew 1997, 1998). There is support from prior research that cross-group friendships are positively associated with positive intergroup attitudes (Turner *et al.* 2007; Davies *et al.* 2011; Feddes *et al.* 2009; Paolini *et al.* 2004, 2007) and that such effects are often achieved by establishing a single cross-group friendship (Raabe and Beelmann 2011; Titzmann *et al.* 2015). Empirical support from prior research leads us to our first hypothesis.

Hypothesis *1: The quantity of face-to-face contact and direct cross-group friendships will be negatively related to prejudice for both communities in Cyprus.*

The impact of extended cross-group friendships (mere knowledge that other ingroup members have outgroup friends) has also been proposed as a very powerful form of indirect contact (Wright *et al.* 1997). In their study, Wright *et al.* (1997) found that White respondents who knew at least one ingroup member with an outgroup friend had weaker outgroup prejudice towards that target group, compared to those who had no extended outgroup friends. Interestingly, they noted that the more extended outgroup friends a participant had, the weaker the prejudice became. None of the comparable previous studies in Cyprus (Husnu

and Crisp 2010; McKeown and Psaltis 2017; Yucel and Psaltis 2020) explored the roles of direct and indirect contact simultaneously and in both communities, as we do in the present paper. These findings from prior research lead to our second hypothesis.

Hypothesis 2: Extended cross-group friendships will be negatively related to prejudice for both communities in Cyprus.

Differential effects of intergroup contact between Greek and Turkish Cypriots and between 2007 and 2017

One critique of the contact hypothesis from prior research is that the positive effect of intergroup contact is not universal. In their meta-analysis of 515 studies, Tropp and Pettigrew (2005) found that there is a stronger effect of intergroup contact among majority groups, where this effect is much weaker among minority groups. However, this argument does not apply that much to the context of Cyprus. In the specific socio-cultural context of Cyprus, the term 'minority' is used for Turkish Cypriots to indicate only the numerical inequality of the two communities but not their political status (McKeown and Psaltis 2017). After the events of 1974, and the resulting complete geographical separation, the situation became even more complicated since the military presence of 30,000–40,000 Turkish troops in the north of Cyprus made many Greek Cypriot Internally Displaced Persons (IDPs) feel under constant threat for returning back to their occupied properties while the general Greek Cypriot population feared further advancement of the Turkish army to the south. The opening of some checkpoints in 2003 was a historical turning point in the Cyprus issue since it reintroduced opportunities for intergroup contact between the two communities. Following the TIMICAT model (see next section), which calls for a more socioculturally situated understanding of contact effects and their inclusion in a time perspective, we discuss intergroup contact in Cyprus and the changing sociopolitical meanings around it, both between Greek and Turkish Cypriots as well as between 2007 and 2017.

The temporally integrated model of intergroup contact and threat (TIMICAT) and the Cyprus context

The Temporally Integrated Model of Intergroup Contact and Threat (TIMICAT) model proposed by Abrams and Eller (2017) addresses the temporal roles of threat and contact in relation to prejudice. The model

proposes that in each societal context, contact and threat have a temporal component, where the effects of contact ('no contact, past contact, discrete single contact, multiple contact, continuous contact, future contact only' (72)) combine with those of threat ('no threat, past threat, discrete single threat, multiple threat, continuous threat, future threat only' (80)). This model is based on two principles. First, different levels of threat or contact at each point in time have a unique impact on prejudice. Each element likewise has its own effect on prejudice. These effects can be cumulative and additive, so that when various temporal forms of threat are at play, this adds to the total impact of the threat. Similarly, the existence of various forms of contact in the temporal frame adds to the total impact of the contact. If contact is positive, then the more contact the better for prejudice reduction. If contact is negative, then this will increase prejudice; the same goes for threats.

The second principle is that the overall impact of contact and threat on prejudice can vary depending on the significant changes in any temporal element. Abrams and Eller (2017) give the example of a cold war, where threats might be pretty stable but the levels of actual contact and opportunities for contact can fluctuate. In this case, the authors expect that the variations in contact would have a stronger direct effect on prejudice. On the other hand, in a case of segregation, where there is little or no contact, then the variations in threat are expected to have a greater role in prejudice. According to Abrams and Eller (2017), the situation becomes complex when different types of threat and contact operate at the same time, such that one or both factors (contact and threat) are either very constant or very variable.

Comparison between 2007 and 2017

Next, we apply the TIMICAT model to the context of Cyprus. This study compares two time periods: 2007 and 2017. Before we make that comparison, though, it is important to give some background information since the effect of intergroup contact can 'differ depending on the events that precede the period of data collection' (Abrams and Eller 2017: 77). In the last 60 years, the situation in Cyprus has gone through various transformations. The period between 1958 and 1974 can be considered as one of *continuous contact but also continuous threat*. During this time, the two communities shared living spaces in more than 100 mixed villages (see Psaltis 2016) and various towns, but these were also the years of intercommunal strife in which many people from both communities were killed,

went missing, or lost properties. The period from 1974 to 2003 can be considered as one of *no contact and moderate or high levels of threat*. During this time, due to complete geographical separation, contact became a memory of the past and a number of discrete threats existed like occasional frictions between security forces across the UN-patrolled zone that separated the two communities. This was a period of minimal bi-communal efforts for reconciliation that mostly took place outside Cyprus. Such a 'cold war' like situation would naturally lead to moderate or high levels of both symbolic and realistic threats, especially given the fact that the two communities have separate educational systems and that for years they largely produced (and consumed) one-sided narratives of victimization.

The 2003 opening of a few checkpoints across the dividing line was a historical turning point that inaugurated a period of partial desegregation, reestablishing some contact between the two communities. It also reflected an increasing number of peace-building activities at the grassroots and NGO levels, along with some serious political efforts to reach a comprehensive settlement under the UN umbrella in both 2004 and 2017. Alongside these positive political efforts, however, there has also been frustration due to the frozen and unresolved nature of Cyprus's longstanding cold war. Despite the different historical reasons, the socio-political context of Cyprus can be predominantly framed as one of moderate to high levels of threat in both communities. Overall, the period since 2003 can be considered as one of *continuous contact but continuous threat.* Although this study does not include measures for threat, the means of the threat scales from previous large-scale representative sample surveys – both from 2007 (Psaltis 2012) and 2017 (Psaltis *et al.* 2020) – reveal high realistic threats, not just symbolic ones, in both communities for 2007 and 2017. While the threat level has been high and stable over the years, according to the TIMICAT model, the variations in actual contact and contact opportunities are expected to have a stronger effect on prejudice.

The decade under examination in this study was a peaceful period of consolidated confidence-building measures (CBMs) and intense efforts to resolve the Cyprus problem. During this time, the number of crossings through checkpoints (and thus contact opportunities) increased.[2] In addition, we expect that with the passing of time, more intergroup friendships will have been established and more people will have learned about these friendships. In addition, reduced symbolic threats and intergroup

[2]An infographic showing the opportunities for contact and crossings can be found here: https://infogram.com/distribution-of-opportunities-for-bi-communal-contact-in-cyprus-cyprob-1hdw2jdxylmj2l0

anxiety are also expected to create more favorable conditions for contact as well as for prejudice reduction. Finally, authorities' increased support of contact, through the work of bi-communal technical committees, is expected to lead to a more favorable environment for contact effects and prejudice reduction in the context of 2017, compared to that of 2007.

Given all these arguments, the effect of the quantity of face-to-face contact on prejudice might be stronger in more recent years (i.e. 2017) compared to earlier years (i.e. 2007). On the other hand, there is evidence for the opposite effect. Specifically, Dhont *et al.* (2011) found among five studies that the effect of intergroup contact on prejudice towards immigrants is higher among those societies in need of closure, which is found to be strongly correlated to racial and gender-based prejudice (Roets and Van Hiel 2011; Van Hiel *et al.* 2004). The argument here is that there is higher potential for the positive effect of intergroup contact on prejudice reduction among those who are highly prejudiced. Along the same lines, given the higher levels of prejudice in 2007 compared to 2017, we could then expect the positive effect of intergroup contact to be stronger in the context of 2007, when there is expected to be a less favorable environment for contact effects and prejudice reduction. Considering these mixed findings and the possibility for opposing arguments, we form the following hypothesis:

Hypothesis 3a: The effect of direct contact on prejudice *will differ between more recent years (i.e. 2017) and earlier years (i.e. year 2007).*

Prior research also shows that the effect of extended contact on outgroup attitudes is stronger when there is no or limited opportunity for direct contact or friendship (Christ *et al.* 2010; Turner *et al.* 2008; Zhou *et al.* 2019). Given the socio-political context and the more favorable environment for any direct contact and higher direct contact opportunities in recent years (i.e. year 2017), as well as for prejudice reduction compared to earlier years (i.e. 2007), we expect that:

Hypothesis 3b: The effect of indirect contact on prejudice *will be stronger in earlier years (i.e. year 2007) compared to more recent years (i.e. year 2017).*

Comparison between Greek and Turkish Cypriots

Developments in the last decade have further been complicated by differences in the way renewed intergroup contact is understood and in internal ideological differentiations within each community (Psaltis 2012), as well

as by varying levels of opportunity for contact due to distance from the nine open checkpoints in the middle of Cyprus. Once the checkpoints were unexpectedly opened by the Turkish Cypriot side in 2003, many people felt it was a situation similar to the fall of the Berlin Wall (according to the TIMICAT model – a *discrete single contact/continuous threat*). Soon it became obvious that there were diverging meanings attributed to the opening in the two communities. Turkish Cypriots saw this as the first step in evaluating whether the two communities can live together again and as a CBM that would facilitate a solution to the Cyprus problem. As a result, in the TC community, about one-third of the population regularly crosses to the south and about 1/3 of TCs have established friendships with GCs, since the TC political parties do not stigmatize crossings and many TCs see crossing as a way of breaking their international isolation. Therefore, for many TCs, the situation has become one of continuous contact. However, given the unresolved nature of the problem, the continuing isolation due to international sanctions, and some discrete threat moments reported in the media (a small number of attacks on TC property when TCs cross to the south of the divide), there is a moderate to high threat context.

On the contrary, the Greek Cypriot leadership and some political parties viewed the opening with suspicion, as an effort to extract recognition for the non-recognized state in the north, which resulted in some political parties actively opposing GCs crossing to the north of Cyprus. One result of this is that about 1/3 of the GCs surveyed have still never crossed to the north. Another third of GCs crossed once or twice to visit lost properties or religious sites, but never visited again (for this group, the context could be described as discrete single contact opportunities under conditions of continuous threat), while the last third cross with some regularity to the north, to develop established intercommunal friendships, to take part in bi-communal civil society organizations, or just to visit places in the north. Previous large-scale representative sample surveys show consistently moderate to high levels of realistic and symbolic threats for both communities since 2007 (Psaltis 2012; Psaltis *et al.* 2020).

Given the almost stable threat level but big variations in the opportunities for contact, it is to be expected that direct contact will gain importance for those having more frequent contact (or more opportunities for face-to-face contact). Prior research has found that quantity of contact has a direct effect on future intentions of contact among TCs (despite their mostly superficial contact with GCs during shopping trips or out of necessity for work), though there was no evidence for the mediating

effect of intergroup trust (McKeown and Psaltis 2017). This could be due to the fact that 'status enhancement' might be a motivation for contact among minority groups (i.e. TCs) (Hagendoorn *et al.* 1998). Due to more frequent crossings by TCs to the south (giving them more contact opportunities with GCs) and the correspondingly few contact opportunities for GCs due to fewer crossings into the north, as well as the much smaller group size of TCs, we form the following hypothesis:

Hypothesis 4a: The effect of quantity of contact on prejudice is expected to be stronger among Turkish Cypriots than Greek Cypriots.

The same study by McKeown and Psaltis (2017) found that the quality of contact is found to affect both outgroup evaluation and future contact intentions among GCs (both effects are mediated by intergroup trust) while there was no such evidence for the quantity of contact among GCs. This finding is consistent with another study (Yucel and Psaltis 2020) which concludes that among GCs, it is the good quality of contact (not the quantity of contact) that affects the willingness for renewed cohabitation with TCs. In addition, while there is less opportunity for face-to-face contact among GCs (due to fewer crossings to the north), deeper and higher-quality interactions (i.e. direct cross-group friendships), along with indirect contact (i.e. extended cross-group friendships), might be more important among GCs. Thus, we formulate the following hypothesis:

Hypothesis 4b: The effects of direct and *extended cross-group friendship on prejudice are expected to be stronger among Greek Cypriots than Turkish Cypriots.*

Data and methods

For the 2007 study, participants who were at least 18-year-old with voting rights were selected by random multistage sampling, by two private research agencies in each community. Face-to-face interviews were conducted at respondents' homes by trained interviewers of the same ethnic origin as the respondent. All data collection took place during February and March 2007, with data gathered from 800 Greek Cypriots and 853 Turkish Cypriots. The 2017 study, which surveyed different respondents from the 2007 study, used original data from GCs and TCs who were at least 18 years old and who had voting rights. The authors of this study prepared the survey questionnaires for each community in

both Greek and Turkish. The University Centre for Field Studies (UCFS) of the University of Cyprus completed a telephone survey of 502 GCs during April 2017. Prologue Consulting Ltd, a research center in North Cyprus, conducted a telephone survey of 600 TCs during September 2017. The fact that these two datasets were collected four months apart might make a comparison harder, due to the collapse of the Cyprus reunification talks in July 2017.[3]

Dependent variables

This study uses prejudice as the dependent variable. Prejudice in both years was measured by asking the respondents to position themselves on a feeling thermometer (Converse and Presser 1986):

> Please rate the GC/TC group on a thermometer that runs from zero (0) to one hundred (100) degrees. The higher the grade, the warmer or more positively you feel towards this group. The lower the degree, the colder or more negatively you feel towards that group.

The answer category ranges from 1 = 0 to 11 = 100.

Independent variables

This study uses three main independent variables: quantity of face-to-face contact, direct cross-group friendships, and extended cross-group friendships. To measure the quantity of face-to-face contact (based on Islam and Hewstone 1993), respondents were asked the following questions for both years:

> Thinking of social contacts (communicating, talking, not just seeing the other person) – whether at home, at work, or somewhere else, how much contact do you have these days with GCs/TCs under the following conditions? (1) at work, (2) in bi-communal meetings, (3) in the area where you live, (4) occasional meetings in the South, and (5) occasional meetings in the North.

The answer categories range from 1=*not at all* to 5=*very often*. The scale is created by averaging these five items. For GCs, Cronbach's alpha for this

[3]In order to make the comparison between these two surveys more valid and reliable, two questions were added to the survey for TCs about whether their thoughts on living together with GCs or having GCs as their neighbors had been influenced by the failed talks. Those who responded yes to either of these two questions were dropped from the analyses, leading to a final sample of 504 TCs. Comparing the complete sample of TCs (N=600) to the final sample we used (N=504), there are no significant differences in terms of our key variables and control variables. This suggests that using the smaller sample (N=504) does not bias our results.

scale is 0.61[4] and 0.78 for years 2007 and 2017, respectively. For TCs, it is 0.77 and 0.88 for years 2007 and 2017, respectively. A higher number on this scale indicates a higher quantity of face-to-face contact. Direct cross-group friendship is measured by asking the respondents the following question (in both years) 'Do you have GC/TC friends?' This question is coded as 1= *yes*, 0=*no*. Extended cross-group friendship is measured by asking the respondents the following question (in both years) 'Does any member of your family have GC/TC friends?'[5] The question is coded as 1=*yes*, 0=*no*.[6]

Control variables

Consistent with prior research, this study also uses several control variables from the survey data: education, age, residence (urban or rural), gender, marital status, religiosity, and year of data collection.

Analytical strategy

This study uses path analysis as part of Structural Equation Modeling (SEM) in AMOS 22. The analyses were first performed separately for

[4]Higher alpha scores indicate higher reliability. Some prior research suggests, however, that 0.60 is the minimum cut-off point for an acceptable alpha (Tabachnick and Fidell 2001; Nunnally and Bernstein 1994). Moreover, the same items for face-to-face contact have been used in prior research on Cyprus (McKeown and Psaltis 2017).

[5]Our data did not include measures on extended contact via less intimate ingroup relationships (i.e., neighbors and work colleagues), as in some prior research (Tausch *et al.* 2011).

[6]Friendships are usually measured in a quantitative form, ranging from none to more than 10 in 5-point Likert scales (Lolliot *et al.* 2015). In fact, the first data set we use (i.e., from year 2007) included such a measure. However, having used this scale to gather data in Cyprus, we found that the scale is highly skewed with the vast majority not reporting any cross-group friendships. For example, according to the 2007 results, 88 percent of GCs and 70 percent of TCs do not have any friends from the other community. In addition, we found that collapsing this scale into a binary form did not distort the findings in any way and better reflected the fact that this scale was essentially capturing the presence or absence of cross-group friendships. For this reason, we decided in our second 2017 study to use the measure in its binary form. Just to make sure that collapsing the measure into a binary would not distort the findings, we compared the zero-order correlations of both measures with the prejudice measure from year 2007. The results were very similar in magnitude (for GCs: r=0.26** on the Likert scale measure and r=0.28** on the binary scale, for TCs: r=0.30** on the Likert scale measure and r=0.25** on the Binary Scale). Additionally, there is longitudinal literature which studied change patterns of intergroup friendships that suggest that the shift from no friendship to some friendship is successful in predicting prejudice reduction, measured in a binary form (Titzmann *et al.* 2015) and that more friendships do not necessarily add something more than a single friendship (Raabe and Beelmann 2011). In addition, for extended cross-group friendship, the measure is also highly skewed in year 2007, with the vast majority of individuals not reporting any extended friendship through family. For example, in year 2007, 90 percent of GCs and 83 percent of TCs do not have any family members who have any friends from the other community. Again, to make sure that collapsing the measure into a binary would not distort the findings, we compared the zero-order correlations of both measures with the prejudice measure from year 2007. The results were very similar in magnitude (for GCs: r=0.22** on the Likert scale measure and r=0.20** on the Binary Scale, for TCs: r=0.21** on the Likert scale measure and r=0.19** on the binary scale). Thus, we decided to measure extended friendship through family in a binary form in year 2017.

years 2007 and 2017 and for Greek and Turkish Cypriots (see Table 2). Model 1 shows the adjusted effects of the quantity of face-to-face contact, direct friendship, and extended cross-group friendship on prejudice for the total sample. Model 2 and Model 3 show the effects of direct and indirect contact for the 2007 and 2017 samples, respectively. Model 4 and 5 show the effects of direct and indirect contact for GC and TC sample respectively. Next, we use multi-group analyses to determine whether the effects of direct and indirect contact vary between Turkish and Greek Cypriots and between the years 2007 and 2017 separately (see Table 3).

Results

Descriptive findings

The full list of variables, along with the descriptive statistics of Greek and Turkish Cypriots and the 2007 and 2017 data, can be found in Table 1. T-tests (and chi-square tests) were performed to test whether there were significant differences in the means (proportions) of the continuous (categorical) variables between Greek and Turkish Cypriots, and between the years 2007 and 2017. Turkish Cypriots have higher prejudice than Greek Cypriots, but the average level of prejudice has decreased significantly between 2007 and 2017. The average quantity of face-to-face contact is the same between Greek and Turkish Cypriots, and this level remained the same between 2007 and 2017. Direct cross-group friendship and extended cross-group friendships are both significantly higher among Turkish Cypriots and both types of friendships increased significantly between 2007 and 2017. In terms of demographic variables, Turkish Cypriots have a younger sample compared to Greek Cypriots, but the average age of the sample has increased significantly between 2007 and 2017. Finally, Greek Cypriots are more religious than Turkish Cypriots, and the average level of religiosity dropped significantly between 2007 and 2017.

Multivariate analyses

Path analysis results

Model 1 shows that for the combined sample, after taking into account the control variables, quantity of face-to-face contact, direct and extended cross-group friendship are all associated with lower prejudice ($b=-0.721$, $p<.001$; $b=-1.173$, $p<.001$, $b=-0.667$, $p<.001$ respectively). Among the combined sample, the results show that Greek Cypriots

Table 1. Descriptive statistics for all variables.

Variable name	Range	GC Sample Mean/Proportion	TC Sample Mean/Proportion	2007 Mean/Proportion	2017 Mean/Proportion
Dependent Variables					
Prejudice	1–11	5.92***(2.58)	7.71***(2.60)	7.32***(2.63)	5.99***(2.71)
Independent Variables					
Quantity of Face-to-Face contact	1–5	1.31 (0.52)	1.32 (0.57)	1.33 (0.51)	1.29 (0.59)
Direct Cross-Group Friendship	0–1	0.15***	0.30***	0.19***	0.28***
Extended Cross-Group Friendship	0–1	0.18***	0.29***	0.16***	0.36***
Control Variables					
Less than high school	0–1	0.24*	0.28*	0.25	0.27
High school graduate or some college degree	0–1	0.51*	0.46*	0.55***	0.38***
University degree or higher (reference)	0–1	0.25	0.26	0.20***	0.35***
Residence (1=Urban, 0=Rural)	0–1	0.72***	0.60***	0.62***	0.71***
Gender (1=Male, 0=Female)	0–1	0.47***	0.57***	0.56***	0.47***
Age	18–92 (GC) 18–89(TC)	49.88*** (16.55)	44.27***(16.21)	42.65***(14.50)	54.40*** (17.32)
Religiosity	1–3	2.61***(0.61)	1.89*** (0.82)	2.35***(0.76)	2.09*** (0.86)

*$p < .05$, **$p < .01$, ***$p < .001$ (two-tailed tests). Standard deviations of continuous variables are in parentheses. The asterisks in the first two columns indicate significant differences between Greek and Turkish Cypriots whereas the asterisks in the last two columns indicate significant differences between years 2007 and 2017.

Table 2. Predictive model of prejudice.

	Total Sample Model 1 b SE	Year 2007 Model 2 b SE	Year 2017 Model 3 b SE	Greek Cypriots Model 4 b SE	Turkish Cypriots Model 5 b SE
Independent Variables					
Quantity of Face-to-Face Contact	−0.721***(0.099)	−0.716***(0.132)	−0.682***(0.151)	−0.309*(0.141)	−1.059***(0.133)
Direct Cross-Group Friendship	−1.173***(0.134)	−1.009***(0.176)	−1.332***(0.205)	−1.435***(0.209)	−0.770***(0.168)
Extended Cross-Group Friendship	−0.667***(0.126)	−0.721***(0.178)	−0.553**(0.176)	−0.765***(0.188)	−0.523***(0.161)
Control Variables					
Less than high school	0.378**(0.142)	0.038(0.189)	0.820***(0.221)	0.227(0.208)	0.426*(0.189)
High school graduate or some college degree	0.280**(0.114)	0.221(0.152)	0.204(0.174)	0.114(0.163)	0.363*(0.154)
Residence	−0.382***(0.099)	−0.346**(0.124)	−0.067396	−0.567***(0.148)	−0.073(0.128)
Gender	−0.165(0.093)	0.000(0.119)	−0.370**(0.147)	−0.084(0.135)	−0.113(0.124)
Age	0.000(0.003)	−0.005(0.004)	0.005(0.005)	−0.033***(0.005)	0.031***(0.004)
Religiosity	0.040(0.065)	0.078(0.084)	0.080(0.111)	0.196(0.111)	0.050(0.081)
Nationality	−2.012***(0.106)	−1.903***(0.128)	−2.175***(0.201)		
Year of data collection	−0.996***(0.108)			−0.539***(0.160)	−1.222***(0.145)
R^2	0.28	0.21	0.33	0.21	0.28

*$p < .05$, **$p < .01$, ***$p < .001$ (two-tailed tests). Standard errors are in parentheses.

Table 3. Fit indices for nested models in multigroup analyses for prejudice.

	COMPARING YEARS 2007 AND 2017			COMPARING GREEK AND TURKISH CYPRIOTS		
Models	df	chi-square	CFI	df	chi-square	CFI
Step 1: the baseline model	8	46.886***	0.99	8	15.451*	0.98
Step 2: the effect of quantity of face-to-face contact is equal across groups	9	46.914***	0.99	9	**30.284*** **	0.98
Step 3: the effect of direct cross-group friendship is equal across groups	9	48.852***	0.99	9	**21.667** **	0.98
Step 4: the effect of indirect cross-group friendship is equal across groups	9	47.337***	0.99	9	16.361	0.98

Note: CFI=comparative fit index; RMSEA= root mean square error of approximation. The chi-squares in bold indicate significant differences compared to the baseline model (i.e., Step 1).
*$p < .05$, **$p < .01$, ***$p < .001$ (two-tailed tests).

have lower prejudice than Turkish Cypriots and respondents report lower prejudice in 2017 compared to year 2007. Lower education is associated with higher prejudice whereas those who live in urban areas report lower prejudice. Using the sample from 2007, the results show that quantity of face-to-face contact, direct and extended cross-group friendship are also associated with lower prejudice (b=−0.716, p<.001; b=−1.009, p<.001; b=−0.721, p<.001). Finally, using the sample from year 2017, quantity of face-to-face contact, direct and extended cross-group friendship are also associated with lower prejudice (b=−0.682, p<.001; b=−1.332, p<.001; b=−0.553, p<.01). Thus, overall, Hypotheses 1 and 2 are supported.

Multi-group analyses

Next, we run multi-group analyses as part of SEM to test whether the effects of direct and indirect contact on prejudice vary between GC and TCs, and between the years 2007 and 2017. Group differences are estimated by comparing a series of nested models (i.e. path coefficients are constrained to be equal across groups) to the baseline model (i.e. path coefficients are freely estimated). Then, using chi-square tests, the model fit between each nested model is compared to the baseline model (Table 3). All the chi-square test results for the comparison between 2007 and 2017 suggest that the difference between the nested model and baseline model is not significant. This suggests that adding equality constraints improve the fit of the baseline model and that the effects of direct and indirect contact on prejudice are the same between the two years (years 2007 and 2017). Thus, Hypotheses 3a and 3b are not supported.

When the same chi-square test results are run for the comparison between the two groups (i.e. GC and TCs), however, two models indicate that constraining some parameters to be equal across groups worsens the fit of the baseline model significantly. Specifically, Step 2 in Table 3 (see bold text) constrains the effect of quantity of face-to-face contact to be equal between Greek and Turkish Cypriots. According to chi-square test results, the effect of quantity of face-to-face contact on prejudice differs significantly between Greek and Turkish Cypriots ($\Delta\chi^2$=14.83, df=1, p<.001), such that the effect is stronger among Turkish Cypriots. Thus, Hypothesis 4a is supported. In addition, Step 3 in Table 3 (see bold text) constrains the effect of direct cross-group friendship to be equal between Greek and Turkish Cypriots. According to chi-square test results, the effect of direct cross-group friendship on prejudice differs significantly between Greek and Turkish Cypriots ($\Delta\chi^2$=6.22, df=1, p<.05), such that the effect is stronger among Greek Cypriots. Thus, Hypothesis 4b is partially supported.

Discussion

In Cyprus, a prime example of geographical separation between two groups with a history of conflict (Greek Cypriots and Turkish Cypriots), the effects of face-to-face contact, (McKeown and Psaltis 2017; Yucel and Psaltis 2020), imagined contact (Husnu and Crisp 2010), online contact through social media friendships (Zezelj *et al.* 2017), and vicarious forms of contact (Husnu *et al.* 2018; Ioannou *et al.* 2018) have been explored in previous research. This study, to our knowledge, is the first one that explores the effects of direct contact (i.e. face-to-face contact and direct cross-group friendship) and extended contact (i.e. extended cross-group friendship) on prejudice, using data from large representative sample surveys outside academic contexts. Moreover, we test whether the effects of direct and indirect contact differ between Greek and Turkish Cypriots, and whether they vary between 2007 and 2017. While a few studies do compare both communities in Cyprus (Yucel and Psaltis 2020; McKeown and Psaltis 2017; Psaltis *et al.* 2019), to our knowledge, this is the first study that tests the effects of intergroup contact in Cyprus by comparing two different time periods. Overall, our study makes a significant contribution to the contact research by considering how and why contextual differences over time might have differential effects of intergroup contact.

The findings reveal no significant difference in both the direct and indirect effects of contact between the two years, despite our prediction that the contacts' effects would differ between years 2007 and 2017. According to the TIMICAT model (Abrams and Eller 2017), the effects of intergroup contact can differ depending on the contextual factors that surround the period of data collection. In our case, the data collection in 2017 coincided with the climax of the effort to resolve the Cyprus problem at Crans Montana in July 2017. At that point, an estimated 25% of people in both communities were clearly against reunification. It is possible that for these people, the Crans Montana efforts elicited a frame of continuous contact combined with a 'merger threat', which temporarily raised symbolic threats and intergroup anxiety, thus reducing the effectiveness of contact on prejudice reduction. It is also possible that in the early years after the opening of the checkpoints, when prejudice levels were significantly higher compared to 2017, that there was more space for positive change (Hodson *et al.* 2017), and that as more years go by and feelings get closer to neutral in both communities, mere contact effects (quantity but not quality) will become weaker.

An interesting pattern is observed when comparing the results of the two communities. In particular, as expected, the quantity of face-to-face contact effect on prejudice reduction is stronger for TCs compared to GCs. In addition, as expected, the significant findings for a stronger effect of direct cross-group friendships on prejudice among GCs (compared to TCs) add support to previous findings from Cyprus relating to the effect of quality contact on prejudice reduction through trust-building (McKeown and Psaltis 2017). In that study, it was reported that among GCs, the quantity of face-to-face contact did not have any effects over and above the quality of contact in increasing trust, which in turn would lead to prejudice reduction. However, in the TC community, there were two direct paths to the outcomes of future contact intentions and outgroup evaluation that were absent in the Greek Cypriot community, which suggested the existence of a more superficial route to prejudice reduction, one that would not entail any change in trust. One path ran from quantity of face-to-face contact to future intentions while the other ran from quality of contact to outgroup evaluations.

Assuming that direct friendships are indicative of qualitative contact, the present finding can be seen as replicating such prior research in Cyprus and confirming the existence of two 'routes' (deep and superficial) for prejudice reduction in TCs but only one for GCs (deep). This finding adds support to the alternative interpretation offered in the previous

paragraph. Namely, as long as mean prejudice levels are high in a community, the mere superficial quantity of face-to-face contact will be much more effective in reducing prejudice than when prejudice levels are moderate. In that case, more qualitative contact would be required to move one from moderate to positive feelings. This would also be in line with the TIMICAT model. The TIMICAT model (Abrams and Eller 2017) distinguishes between positive and negative contact in testing the temporal effects of contact and threat on prejudice. Thus, in order to better evaluate the TIMICAT model, it would be beneficial for future research to also include measures of negative and positive contact in considering the effect of intergroup contact.

In the case of TCs, given the official master narrative of rejection by GCs and marginalization, mere contact which disproves such negative expectations (e.g. going shopping in the south without an instance of negative contact) could be enough of a socio-cognitive conflict to change the representation of the outgroup. In the case of GCs, the official master narrative is one of having 'no problem with TCs but with the Turks' and that 'we lived together in peace in the past so we can do it again'. In this case, superficial contacts would not create much of a socio-cognitive conflict, especially for older people who have lived together with TCs in the past. However, good quality contact is always expected to lead to trust, creating a position to lead a person with moderate feelings towards more positive feelings. Given the differential effects of different forms of contact for Greek Cypriots and Turkish Cypriots, there is a need for further exploration of a possible dual-process model of contact effects (Moscovici 1976).

Nevertheless, this study also features findings that differ from previous research, in terms of the effects of quantity of face-to-face contact over and above any effects of quality of contact on prejudice reduction. In this study, we find that more contact (controlling for quality of contact) leads to prejudice reduction in both communities, but even more for TCs. Given other findings (Yucel and Psaltis 2020) that contact effects in Cyprus are moderated by age, it is possible that the present positive findings of quantity of face-to-face contact are driven by the older TC sample, which is more prejudiced compared to the younger TC sample. The fact that our sample is representative means that it includes a larger age range in contrast to the McKeown and Psaltis (2017) study which mainly featured a younger student sample.

Generally speaking, in both communities we found evidence that extended cross-group friendships and number of friendships increased

significantly between 2007 and 2017, as expected, which is one of the more important findings of the present study. This increase in extended cross-group friendships is related to an increase in the percentage of individuals in both communities who have cross-group friendships. As expected, in small and tightly-knit communities like the two Cypriot communities, the visibility and impact of such cross-group friendships are multiplied by extended forms of contact (see Granovetter 1973). Overall, the present findings suggest that given the low quantity of face-to-face contact in Cyprus, cross-group friendships and extended cross-group friendships gain importance. The findings in this study are consistent with the original contact hypothesis (Allport 1954), the meta-analyses findings by Pettigrew and Tropp (2006), and the more recent meta-analysis of extended contact effects by Zhou *et al.* (2019). The quantity of face-to-face contact, direct friendship, and extended cross-group friendship are all associated with lower prejudice.

There are several limitations to this research. First, the data we use has a repeated cross-sectional design where data is obtained from different samples from the same population at two different time points. Thus, we cannot show any causal relationships between the key variables, so we present the correlations instead. Second, the changes identified between years 2007 and 2017 could be only be partly attributed to direct and indirect forms of contact. There is a need for future research to explore the changing role of economic or other forms of threats (such as realistic and symbolic threats) in relation to sociopolitical developments over the years (Abrams and Eller 2017), preferably using multilevel analysis or time series analysis considering the macro-level variables (Ron *et al.* 2017). Third, the method of data collection was different in years 2007 and 2017. In addition, given the different forms of extended contact via intimate ingroup relationships (i.e. family and friends) and less intimate ingroup relationships (i.e. neighbors and work colleagues) (Tausch *et al.* 2011), future research could include these various forms of extended contact and test whether there are differences in their effects on the outcomes of interest. Finally, while we explained earlier why we used a binary scale to measure direct and extended cross-group friendships, rather than the continuous measure we used for quantity of face-to-face contact, these different scales do not allow us to test the relative importance of each type of contact on prejudice, but rather the individual effects of each on prejudice. Future research, with comparable measures of both direct and indirect contact, would be able to empirically compare their effects on prejudice.

There are certain policy implications of these results for the particular context of Cyprus. Previous research in Cyprus identified the beneficial effects of direct face-to-face contact in separate studies in the GC and TC communities (McKeown and Psaltis 2017), but this was the first study to compare results from two different points in time for both communities simultaneously. Given its large-scale representative sample, this study provides the most comprehensive test of comparative forms of Cypriot contact to date. The quantity of intergroup contact, and especially its quality (more for GCs than TCs), is clearly of crucial importance for prejudice reduction, an outcome of special importance in the efforts towards reconciliation and renewed cohabitation of the two communities under power-sharing arrangements. The present findings also highlight the special importance of extended forms of contact in a context of high geographical segregation which nevertheless offers a lot of opportunities for perceived extended contact, given the small size of the communities and the region's tightly-knit nature. Moreover, given the changes over time demonstrated in this study, our results highlight the importance of context in understanding intergroup contact, and how the effect of intergroup contact changes across contexts.

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